Palm OS Programming Recipes

Recipe: Initializing Text in a Simple Editable Field

The most common use of fields is to take user input. Using a field in this way is very easy - the system takes care of allocating, resizing and freeing memory for you, though you can handle the memory allocation or freeing if necessary.

To create an editable field:

1. Place your field on your form, sized as you like it. Make sure not to make the field shorter than the font's line height, otherwise it might not display at all.
2. The field should be "Usable", "Editable", and "Underlined".
3. You may or may not want it to be Single Line. Note that if the user enters a text string that is visibly longer then the width of the single line field, the field will not scroll such that they will not see what they are writing anymore.
4. You probably do not want it to be "Dynamic Size". That is used when the field needs to grow vertically when the user enters enough text; you have to handle fldHeightChangedEvent to make it work.
5. Set "Max Characters" to the most characters you want in the field. If you are using a "Single Line" field, you should set the Max Characters such that any reasonable entry will not be wider then can display on a line. Try typing in capitol 'M' until you run off the edge for the worse case scenario.
6. Optionally set "Auto Shift" or "Numeric" if you're certain you know what sort of data is going to be entered in this field.
7. For this example, don't set "Has Scroll Bar".

If you add no further code at all, the field will accept user input, and will allocate and resize memory as appropriate to hold the user's input, and will free that memory when the form is closed - in other words, by default you need to do nothing but read the user's input at the appropriate time by calling FldGetTextHandle or FldGetTextPtr, and perhaps FldCompactText first. (Make sure to copy out the text, though, since the pointer might well change if the user enters more text.)

Also, do not modify the data in the string directly, unless you again call FldSetTextHandle or FldSetTextPtr immediately after doing so. Fields keep some internal information about the string, such as its length, where line breaks are, and so forth, so if you change the text without telling the field you may well get bad results or error messages when the field notices that something is wrong.

To initialize the text in a field use the following:

```
MemHandle h, oldHandle;
char *p;
FieldType* fldP;
FormType* frmP;

frmP = FrmGetActiveForm();
h = MemHandleNew(50);
p = (char *) MemHandleLock(h);
StrCopy(p, "George Washington");  // you'd copy from saved data, presumably
MemHandleResize(h, StrLen(p)+1);
```

```
MemHandleUnlock(h);
fldP = (FieldType *) FrmGetObjectPtr(frmP, FrmGetObjectIndex(frmP, MainNameField));
oldHandle = FldGetTextHandle(fldP);
FldSetTextHandle(fldP, h);
if (oldHandle)
   MemHandleFree(oldHandle); // or do something else with it first
```

If you don't want the field to free the handle when the form closes, then in your frmCloseEvent get the handle with FldGetTextHandle() and then set the field to use a zero handle with FldSetTextHandle(fldP, 0). Then you are the "owner" of the handle containing the user's input, and can use it as you wish, remembering to free it when you are done.